Mohamed Ahmed Ali

Cairo, Egypt, 01124340812, mohamed.ahmed159840@gmail.com

LINKS

Linkedin, Github, Portfolio

PROFILE

Mechatronics Engineer with extensive experience in Backend Development, Data Analysis, Product Ownership, Testing, and Embedded Software & Robotics. Proven track record in designing and implementing solutions that bridge technical and business requirements. Enthusiast of AI technologies and passionate about game development. Holding a Bachelor's degree in Mechatronics Engineering. Seeking a challenging position to leverage my diverse engineering background in a dynamic and innovative environment.

EMPLOYMENT HISTORY

Jul 2025 — Present

Backend Developer, Imdova — Cairo

- Developed the backend infrastructure for the **Courses platform** serving undergraduate medical students with on-demand and live training content.
- Built scalable RESTful APIs to manage user accounts, course enrollment, video content, assessments, and certification features.

Apr 2025 — Present

Backend Developer, Spotter — Cairo

- **Contributed to the development of a fitness coaching platform** that offers personalized advice, progress tracking, and AI-powered chatbot interactions to help users achieve their health goals.
- **Ensured clean architecture** through modular service design and scalable authentication strategies.

Jan 2024 — Jul 2025

NestJs Backend Developer, Arcon Corporation — Cairo

- **Designed and developed scalable RESTful APIs** for large-scale applications, including educational and healthcare platforms.
- **Implemented role-based access control (RBAC)** to manage user permissions securely across different roles.
- **Developed and executed integration tests** to ensure modules interact seamlessly and meet business logic requirements.
- **Integrated Firebase Authentication**, streamlining secure user management.

Sep 2022 — Dec 2023

Product Owner, Arcon Corporation — Cairo

- **Bridged communication between business and development teams**, translating project requirements into actionable tasks.
- **Developed and executed SQL-based data analysis**, providing insights into user behavior and platform performance to drive data-driven decisions.

Sep 2021 — Aug 2022

Embedded Software & Robotics Engineer, Arcon Corporation — Cairo

- **Developed and integrated embedded systems** for **51+ scientific projects**, utilizing Arduino microcontrollers, sensors, and electronic components.
- **Led robotics development and supervised VEX robotics competitions**, mentoring teams and optimizing performance strategies.

EDUCATION

Sep 2016 — Jul 2021

Bachelor of Science Mechatronics Engineering, Faculty of Engineering Ain Shams University — Cairo

Graduated with the degree of Very Good (2.98 GPA)

Graduation Project Grade: A+

SKILLS

NestJS	Experienced	JAVA	Skillful
SQL	Experienced	C, C++	Skillful
TypeScript	Experienced	JavaScript	Skillful
Google Firestore	Experienced	Machine Learning	Experienced
BigQuery	Experienced	Laravel	Expert

Backend APIs	Skillful	Power BI	Expert
UX/UI	Beginner	Postgres	Expert
Python	Experienced	PHP	Skillful

PROJECTS

Mar 2022 — Present

LMS System

This LMS, built with **NestJS**, **Firebase Authentication**, and **PostgreSQL**, supports role-based access control for admins, teachers, and students, enabling efficient learning management.

Roles and Functionalities

- **Admin**:Manage users (add/edit/delete teachers and students).
- **Teacher**:Create courses and classes, assign students, and manage assignments. Monitor and reset student scores.
- **Student**:Access assigned courses, submit assignments, and participate in experiments.

Technical Highlights

1. **NestJS**: Modular and scalable backend framework.
2. **Firebase Authentication**: Secure, role-based access control.
3. **PostgreSQL**: Manages users, courses, assignments, and relationships.

Link

Sep 2024 — Oct 2024

NestJS Multi-Role Application with MongoDB

Developed a scalable NestJS application with MongoDB, supporting three roles: Admin, Vendor, and Consumer. Key features include:

- **Admin:** Manage users with CRUD operations.
- **Vendor:** Manage products and inventory.
- **Consumer:** User registration, authentication, and cart management.

Highlights:

- **Secure Authentication & Authorization:** JWT-based with role-based access control.
- **RESTful APIs:** Comprehensive endpoints for users, products, and carts.
- **Scalable Architecture:** Modular and extensible design adhering to best practices.

Link

Sep 2024 — Sep 2024

Real-Time Chat Application with NestJS and Vue

Built a real-time chat application with a NestJS backend using WebSockets and a Vue.js frontend. The app enables seamless, interactive communication between multiple users, featuring:

- **Real-Time Messaging:** Instant updates across clients when messages are sent.
- **Typing Indicators:** Shows when a user is typing to enhance interactivity.
- **Scalable Backend:** WebSocket-based communication powered by NestJS for efficient message handling.
- **Responsive Frontend:** Vue.js provides a clean and intuitive interface for conversations.

Link

Sep 2021 — Present

Siminds LabsForHome

A cutting-edge 3D simulation platform for scientific experiments in physics, chemistry, and biology, designed to enhance interactive and accessible STEM education. The system incorporates the Arduino microcontroller, I2C LCD, and a variety of sensors to simulate and visualize real-world experiments.

Key Features:

- **Comprehensive Experiment Library:** Offers over 51 detailed experiments in both Arabic and English, catering to diverse educational needs.
- **Accessibility-Focused Game Modes:** Includes innovative features such as eye-tracking and voice commands to ensure inclusivity for individuals with disabilities.
- **Realistic 3D Models:** High-detail visualizations provide an immersive learning experience, replicating laboratory setups with precision.
- **Interactive Avatars:** Virtual assistants guide students through experiments, offering step-by-step support and feedback.
- **Seamless Hardware Integration:** Real-time synchronization with physical Arduino kits enables users to input and analyze actual experimental data within the simulation.

Link

Jun 2022 — Jun 2022

Unity Hand Tracking

- Implemented real-time hand landmark tracking in Unity for robotic control.
- Enhanced human-robot interaction (HRI) for potential humanoid robot applications.

Link

Sep 2023 — Sep 2023

Controlling Arduino Using Python Hand Tracking

- Built a system to control robotic actuators using hand-tracking data.
- Developed a Python-based gesture recognition system to interface with Arduino-controlled devices.

Link

Jul 2018 — Present

Mega Mania

Developed a complete Java platformer game featuring various levels, power-ups, and enemies.

Link

Jan 2021 — Jan 2021

Flappy Bird Using Machine Learning

- Built a self-playing AI agent using neural networks and genetic algorithms.
- Trained the model to optimize decision-making based on reinforcement learning.

Link

Nov 2023 — Dec 2023

DOOM Clone

Programmed a DOOM clone in Python, featuring different weapons, enemies, and levels.

Link

Oct 2020 — Jul 2021

Soft Robotic Glove For Rehabilitation

- Designed a wearable robotic glove using flex sensors and microcontrollers.
- Assisted patients with rehabilitation by tracking finger movement and force feedback.
- Implemented AI-assisted movementrecognition for improved rehabilitation exercises.

Link

May 2022 — May 2022

Chatbot

Created a Python-based chatbot capable of learning and responding to user queries.

Link

Feb 2020 — Jun 2020

Lotfy The Butler Robot

Designed a mobile robot butler using object detection and GPS technology to assist with luggage carrying during travel.

Link

CERTIFICATIONS

IoT Development Using M.O.T Platform (ITI MaharaTech)

Robotics Workshop (iHub)

Software Testing (ITI MaharaTech)

Gamification in eLearning (ITI MaharaTech)

UX Design Fundamentals (ITI MaharaTech)